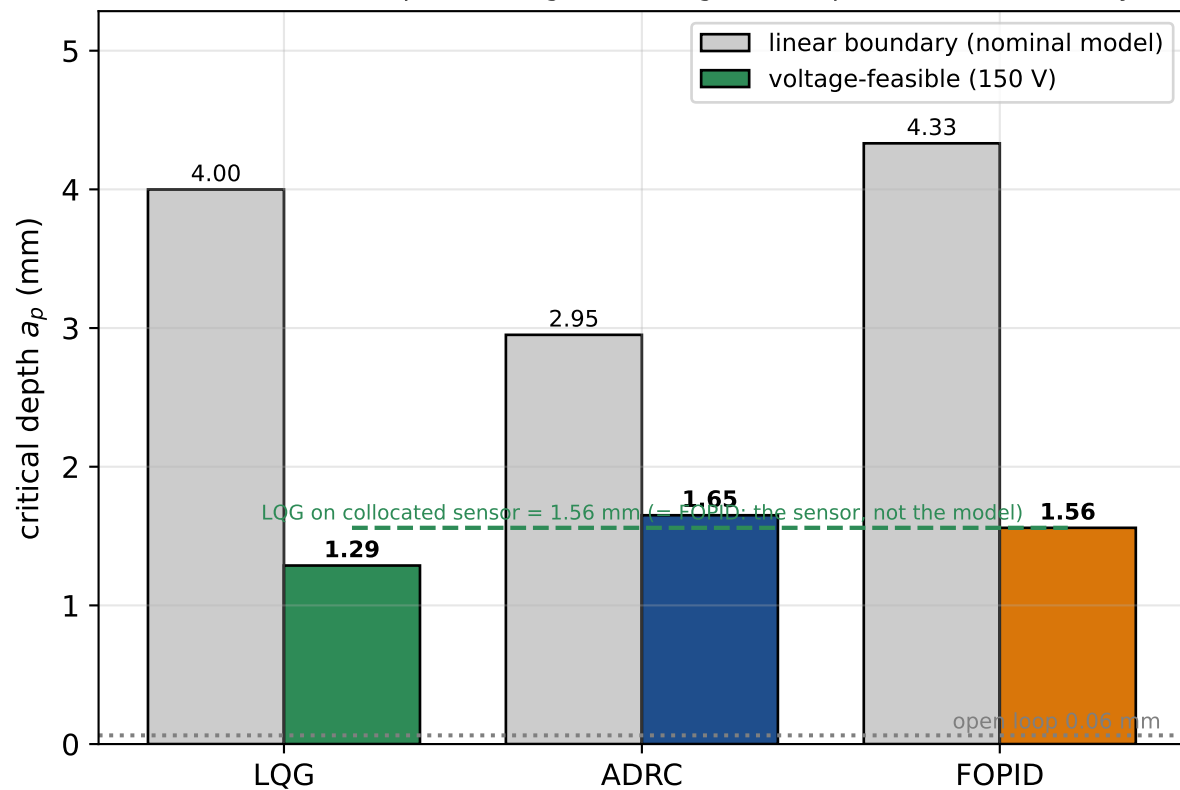
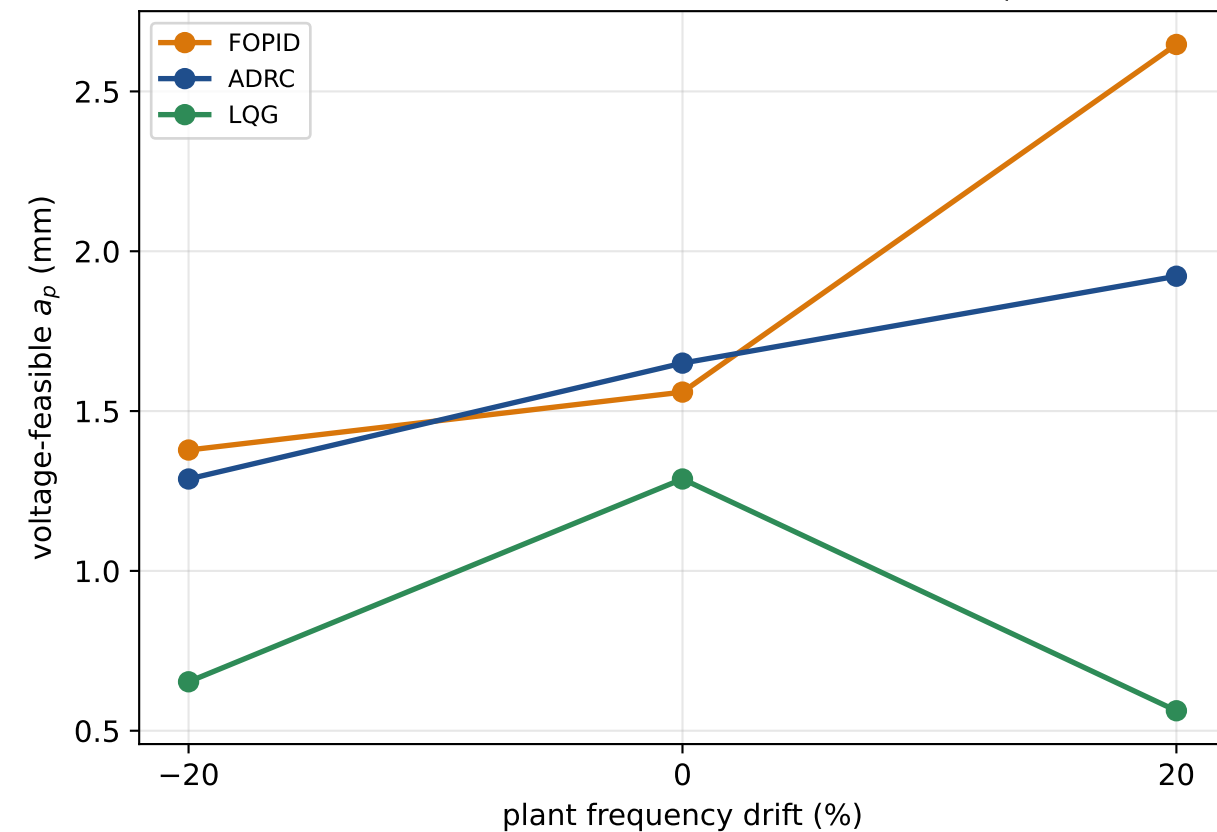


Fractional-order PID (FOPID) vs LQG / ADRC — identical plant, sensors, saturation and metrics

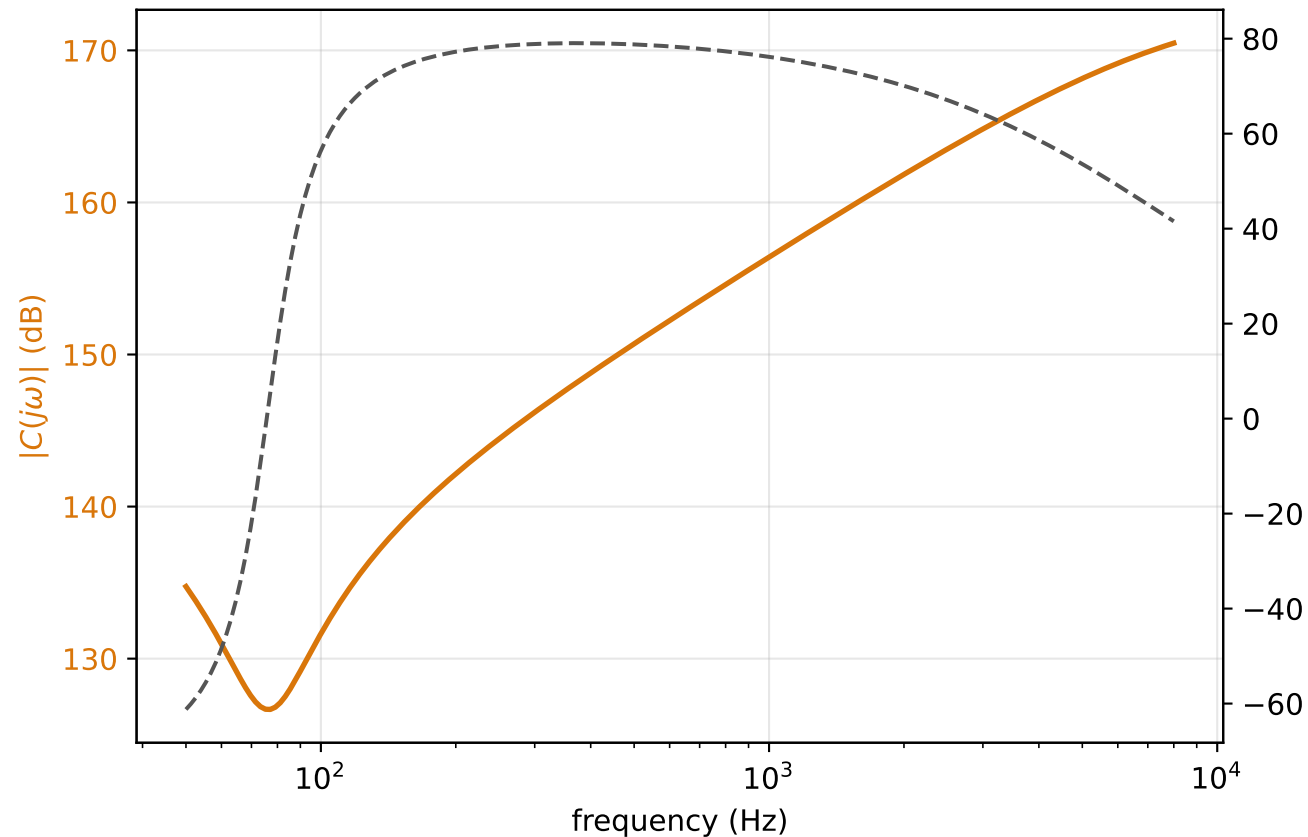
(a) linear boundary (nominal) vs voltage-feasible depth
FOPID = LQG on equal sensing; the voltage wall caps the linear boundary



(b) robustness: feasible depth under frequency drift
collocated FOPID/ADRC hold; model-based LQG collapses



(c) designed FOPID $C(s) = K_p + K_i s^{-0.94} + K_d s^{0.92}$
(dashed = phase; fractional slopes)



(d) tip response at $a_p = 1.2$ mm
ADRC rejects; FOPID stabilises but $\sim 10\times$ more residual

